# ANUPAM SINGH

**Full-Stack Developer — MERN · NestJS · React**

anupamsingh2389@gmail.com | 70651-69819 | India | Github | Linkedin | Portfolio

## SUMMARY

Full-Stack Developer with 4+ years building and operating production systems on Node.js, NestJS, React, and AWS. Owns backend architecture end-to-end — multi-provider integrations, subscription billing, revenue attribution, and distributed systems — on a platform serving 100K+ registered students and 220K+ submitted applications. Drives design decisions, deployment, and observability across the full delivery lifecycle.

## EXPERIENCE

**Lenny.ai** | Israel — Remote **May 2023 – Present**

*Software Engineer (Full Stack) — LENNY.AI scholarship-management platform & BERTI.AI (multi-model AI platform)* **Platform**

**scale:** 100K+ registered students, 220K+ scholarship applications submitted through the platform.

### LENNY.AI — SCHOLARSHIP PLATFORM (LINK)

- Engineered a **20+ step branching signup funnel** (SignupV2) for LENNY.AI's scholarship platform — conditional scholarship-offer variants, pricing-bucket resolution, and end-to-end referrer/UTM attribution — owning the platform's **highest-traffic conversion path**.
- Built LENNY.AI's scholarship-management platform end-to-end — a CRUD + publishing pipeline for scholarships, an applicant review workflow, and a configurable **rank-criteria/scoring system** for winner selection — plus a dynamic, multi-theme application form builder; remediated an **XSS vulnerability** via DOMPurify sanitization across **10+ components**.
- Designed a **partner/affiliate revenue-attribution system** from scratch — postback endpoints, encrypted partner API keys, click-to-conversion tracking — integrated with **Meta Conversions API** (server-side fbc/fbp capture) and GA4; instrumented **150+ frontend components** across Mixpanel, PostHog,clarity, and Hotjar; synced lead lifecycle via a **Mautic** email-automation integration.
- Built **multi-tenant SaaS routing** for a single React SPA serving multiple client organizations off one codebase, **eliminating the need for per-tenant deployments**; centralized JWT auto-injection and error normalization across **100+ API call sites** via a shared Axios interceptor layer.

### BERTI.AI — MULTI-MODEL AI PLATFORM (LINK)

- Architected a **multi-provider LLM orchestration layer** (NestJS) unifying **8+ AI providers** (OpenAI GPT, Anthropic Claude, Google Gemini, DeepSeek, xAI Grok, Perplexity, Stability AI/Leonardo AI) for BERTI.AI, plus the **side-by-side multi-model comparison UI** (React 18, Redux Toolkit) letting users query up to **3 AI models simultaneously** — the product's core differentiator.

### PLATFORM-WIDE

- Built and maintained **multi-gateway subscription billing** across **5 payment providers** (PayPal, Stripe, CheckoutChamp, Paddle, ArcaMax) with OAuth2 client-credentials flows and webhook-driven reconciliation; designed a coupon/voucher engine and a tier-based usage-metering system.
- Designed and shipped the core **REST API surface** for the platform on a Fastify-based NestJS backend, standardizing auth and access control via global JWT middleware, Google OAuth2, and CASL-based RBAC down to individual features — fully documented via Swagger/OpenAPI.
- Owned AWS deployment — **EC2, PM2, Nginx** on the backend and **S3 + CloudFront + Route 53** on the frontend, with transactional email delivery via **AWS SES**.

**Welkin Ring Digital Solutions** | Faridabad **Apr 2022 – Apr 2023**

*JavaScript Developer (MERN)*

- Built secure, scalable backend systems (Node.js, Express, MySQL) with an integrated HDFC payment gateway; migrated legacy PHP systems to Node.js and optimized MySQL queries/indexing, **reducing load times by 30%**.

## PROJECTS

**Blogify — Full-Stack Microservices Blog Platform (GITHUB LINK) (DEMO)**

*React 19, Node.js/Express, MongoDB, PostgreSQL, RabbitMQ, Redis, Docker, AWS*

- Architected a production microservices platform of **4 independently deployable services** (API gateway, user, blog, AI) with strict per-service data ownership across MongoDB and PostgreSQL and **zero cross-service database joins**, communicating via REST, a **RabbitMQ event backbone (3 durable topic exchanges)**, and real-time WebSockets.
- Built stateless **JWT auth with refresh-token rotation and reuse detection**, Redis-backed instant revocation, and the **circuit breaker pattern** (Opossum) with a fail-open strategy across Redis/RabbitMQ, **so infrastructure outages degrade gracefully instead of cascading into full failures**; containerized a **12-container stack** via Docker Compose with GitHub Actions CI/CD and full OpenTelemetry/Prometheus/Loki/Tempo observability.

**NeoDrive — Cloud File Storage Platform (GITHUB LINK) (DEMO)**

*React, Node.js/Express, MongoDB, AWS (S3/CloudFront), Redis, BullMQ, Docker, OpenTelemetry*

- Architected a full-stack cloud storage platform supporting hierarchical folders, **direct-to-S3 uploads/downloads** (presigned PUT + CloudFront-signed GET), quota-managed storage, and Razorpay billing.
- Engineered a two-phase upload pipeline with atomic MongoDB compare-and-increment quota reservations, eliminating a **race condition**; built independent**, path-filtered CI/CD pipelines** (GitHub Actions) deploying frontend and backend separately with **zero cross-pipeline interference.**

## SKILLS

**Primary:** TypeScript, JavaScript, Node.js, NestJS, Express, React, PostgreSQL, MongoDB

**Infrastructure:** AWS (EC2/S3/CloudFront), Docker, Nginx, Redis, RabbitMQ, GitHub Actions (CI/CD)

**Architecture:** REST, WebSockets, Microservices, OAuth2, JWT, RBAC, Payment Gateways (PayPal, Stripe, Razorpay)

**Observability:** OpenTelemetry, Prometheus, Grafana, Loki, Tempo, Jaeger

## EDUCATION

Gurugram University — MCA | Sep 2023 – May 2025 Maharshi Dayanand University, Rohtak — BCA | Aug 2019 – Jun 2022